



12764 - Zooming In on SN2026ngr: A Gravitationally Lensed Type Ia Supernova Magnified by 100×

Cycle: 4, Proposal Category: DD

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
spectroscopy				
	2	IFU SN2026ngr	NIRSpec IFU Spectroscopy	(3) SN2026ngr
	3	IFU SN2026ngr	NIRSpec IFU Spectroscopy	(3) SN2026ngr
	1	IFU SN2026ngr	NIRSpec IFU Spectroscopy	(3) SN2026ngr
NIRCam 1				
	4	Imaging SN2026ngr	NIRCam Imaging	(3) SN2026ngr
	5	Imaging SN2026ngr	NIRCam Imaging	(3) SN2026ngr

ABSTRACT

The Zwicky Transient Facility has recently discovered SN2026ngr, a strongly gravitationally lensed Type Ia supernova at redshift $z=0.95$. Its observed brightness implies an extraordinary lensing magnification of approximately 100x, making it a unique laboratory for both cosmology and supernova physics. We propose three epochs of JWST/NIRSpec IFU spectroscopy to exploit the rapid spectral evolution of Type Ia supernovae as an independent clock, enabling precise measurements of the relative phases of the multiple SN images and, consequently, accurate determinations of the lensing time delays. Combined with high-resolution NIRCam imaging, these observations will provide the astrometric precision, image flux ratios, colors, and host-galaxy arc morphology required for robust lens modeling and a measurement of the Hubble constant, H_0 , at an intermediate redshift between the cosmic microwave background and local distance-ladder probes.

Beyond its cosmological value, the extreme magnification transforms this system into a powerful gravitational telescope. The proposed spectroscopic and photometric observations will deliver an unprecedented view of a Type Ia supernova at $z\sim 1$, allowing detailed measurements of key spectral diagnostics, including the Si II and Ca II infrared features, and a direct comparison with their low- z counterparts. These observations will provide a rare opportunity to test the robustness of Type Ia supernova standardization at cosmologically relevant redshifts and help establish the observational foundations for the large samples of strongly lensed supernovae expected from Rubin and Roman in the coming decade.

OBSERVING DESCRIPTION

We propose observations of the newly discovered strongly lensed Type Ia supernova, SN 2026ngr at $z=0.95$. The proposed plan includes 3 visits with NIRSpec/IFU and 2 with NIRCam, with about 2 weeks cadence, all to be completed before the window of observability expires by the end of July 2026.

Proposal 12764 - Targets - Zooming In on SN2026ngr: A Gravitationally Lensed Type Ia Supernova Magnified by 100×

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(3)	SN2026ngr	RA: 13 58 24.3643 (209.6015179d)	Proper Motion RA: 0	
			Dec: -03 02 48.86 (-3.04691d)	Proper Motion Dec: 0	
			Equinox: J2000	Epoch of Position: 2000	
	Comments: Category=Star Description=[Type Ia supernovae]				

Proposal 12764 - Observation 2 - Zooming In on SN2026ngr: A Gravitationally Lensed Type Ia Supernova Magnified by 100×

Observation	Proposal 12764, Observation 2: IFU SN2026ngr											Tue Jun 16 18:03:45 GMT 2026
	Diagnostic Status: Warning											
Diagnostics	Observing Template: NIRSpec IFU Spectroscopy											
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(3)	SN2026ngr	RA: 13 58 24.3643 (209.6015179d) Dec: -03 02 48.86 (-3.04691d) Equinox: J2000				Proper Motion RA: 0 Proper Motion Dec: 0 Epoch of Position: 2000					
Template	Comments: Category=Star Description=[Type Ia supernovae]											
	TA Method						HFF Readout Mode					
Dithers	NONE						false					
	#	Dither Type		Size		Starting Point		Number of Points		Points		
Spectral Elements	1	4-POINT-DITHER										
	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
Special Requirements	1	G140M/F100LP	NRSIRS2RAPID	85	1	false	true	NONE	4	4	5018.578	27082
	2 After 1 by 10 Days to 20 Days 3 After 2 by 10 Days to 20 Days											

Proposal 12764 - Observation 3 - Zooming In on SN2026ngr: A Gravitationally Lensed Type Ia Supernova Magnified by 100×

Observation	Proposal 12764, Observation 3: IFU SN2026ngr												Tue Jun 16 18:03:45 GMT 2026
	Diagnostic Status: Warning												
Diagnostics	Observing Template: NIRSpec IFU Spectroscopy												
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(3)	SN2026ngr	RA: 13 58 24.3643 (209.6015179d) Dec: -03 02 48.86 (-3.04691d) Equinox: J2000				Proper Motion RA: 0 Proper Motion Dec: 0 Epoch of Position: 2000						
Template	Comments: Category=Star Description=[Type Ia supernovae]												
	TA Method						HFF Readout Mode						
Dithers	NONE						false						
	#	Dither Type		Size		Starting Point		Number of Points		Points			
Spectral Elements	1	4-POINT-DITHER											
	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
Special Requirements	1	G140M/F100LP	NRSIRS2RAPID	85	1	false	true	NONE	4	4	5018.578	27082	
	3 After 2 by 10 Days to 20 Days												

Proposal 12764 - Observation 1 - Zooming In on SN2026ngr: A Gravitationally Lensed Type Ia Supernova Magnified by 100×

Observation	Proposal 12764, Observation 1: IFU SN2026ngr												Tue Jun 16 18:03:45 GMT 2026
	Diagnostic Status: Warning												
Diagnostics	Observing Template: NIRSpec IFU Spectroscopy												
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(3)	SN2026ngr	RA: 13 58 24.3643 (209.6015179d) Dec: -03 02 48.86 (-3.04691d) Equinox: J2000				Proper Motion RA: 0 Proper Motion Dec: 0 Epoch of Position: 2000						
Template	Comments: Category=Star Description=[Type Ia supernovae]												
	TA Method						HFF Readout Mode						
Dithers	NONE						false						
	#	Dither Type		Size		Starting Point		Number of Points		Points			
Spectral Elements	1	4-POINT-DITHER											
	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
Special Requirements	1	G140M/F100LP	NRSIRS2RAPID	85	1	false	true	NONE	4	4	5018.578	27082	
	Before Date 25-JUN-2026:00:00:00 2 After 1 by 10 Days to 20 Days												

Proposal 12764 - Observation 4 - Zooming In on SN2026ngr: A Gravitationally Lensed Type Ia Supernova Magnified by 100×

Observation	Proposal 12764, Observation 4: Imaging SN2026ngr									Tue Jun 16 18:03:45 GMT 2026
	Diagnostic Status: Warning Observing Template: NIRCam Imaging									
Diagnostics	(Imaging SN2026ngr (Obs 4)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified. (Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous	
	(3)	SN2026ngr	RA: 13 58 24.3643 (209.6015179d) Dec: -03 02 48.86 (-3.04691d) Equinox: J2000			Proper Motion RA: 0 Proper Motion Dec: 0 Epoch of Position: 2000				
	Comments: Category=Star Description=[Type Ia supernovae]									
Template	Module			Subarray			Target Placement			
	ALL			FULL			Module gap (large extended source)			
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	INTRAMODULEBOX		4		STANDARD				1
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F150W	F277W	BRIGHT1	6	1	4	4	472.418	
	2	F115W	F277W	BRIGHT1	6	1	4	4	472.418	
	3	F090W	F277W	BRIGHT1	6	1	4	4	472.418	
Special Requirements	Between Dates 19-JUN-2026:00:00:00 and 11-JUL-2026:00:00:00 Offset 121.0 arcsec, 35.0 arcsec 5 After 4 by 10 Days to 30 Days									

Proposal 12764 - Observation 5 - Zooming In on SN2026ngr: A Gravitationally Lensed Type Ia Supernova Magnified by 100×

Observation	Proposal 12764, Observation 5: Imaging SN2026ngr									Tue Jun 16 18:03:45 GMT 2026
	Diagnostic Status: Warning									
Observing Template: NIRCam Imaging										
Diagnostics	(Imaging SN2026ngr (Obs 5)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified.									
	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous	
	(3)	SN2026ngr	RA: 13 58 24.3643 (209.6015179d)			Proper Motion RA: 0				
			Dec: -03 02 48.86 (-3.04691d)			Proper Motion Dec: 0				
			Equinox: J2000			Epoch of Position: 2000				
Template	Comments: Category=Star Description=[Type Ia supernovae]									
	Module		Subarray				Target Placement			
	ALL		FULL				Module gap (large extended source)			
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	INTRAMODULEBOX		4		STANDARD				1
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F150W	F277W	BRIGHT1	6	1	4	4	472.418	
	2	F090W	F277W	BRIGHT1	6	1	4	4	472.418	
	3	F115W	F277W	BRIGHT1	6	1	4	4	472.418	
Special Requirements	Offset 121.0 arcsec, 35.0 arcsec									
	5 After 4 by 10 Days to 30 Days									